NanoTabPFN Pretraining Speedrun

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TL;DR

modded-nanoTabPFN: This repository hosts the nanoTabPFN speedrun, in which we search for the fastest way to train a tabular foundation model that beats Random Forest on TabArena datasets.

Introduction

Motivation

- TabPFN have shown that pretraining on synthetic datasets can lead to strong performance... [1] [2]
- However, training these models takes long, and no one has time to wait...

Background

- GPT2 → nanoGPT → modded-nanoGPT (45 mins to 1.54 mins)
- TabPFNv2 → nanoTabPFN → modded-nanoTabPFN

Goal

- Pretrain a neural network to beat Random Forest (~0.8068 validation average ROC AUC) on subsampled TabArena datasets using 1 NVIDIA L40S.

This repo now contains a training algorithm which attains the target performance in:

- **0.92 minutes** on 1xL40S (baseline needed 74.32)
- **3648 synthetic datasets** (baseline needed 80576)

Rules

New records must:

1. Not modify the evaluation pipeline.
2. Not load any pretrained weights.
3. Run faster than prior record when baselined on the same hardware.

Other than that, anything and everything is fair game, including changes to the synthetic prior generation!

Timing protocol

- Counting only training wall-clock time (forward/backward/optimizer steps over synthetic batches).
- Evaluation runtime is excluded.
  - the training timer is stopped while running validation.
- Prior generation time is excluded.
  - using a pre-generated prior dump is allowed.

Evaluation details

Evaluation is on all 38 TabArena classification datasets:

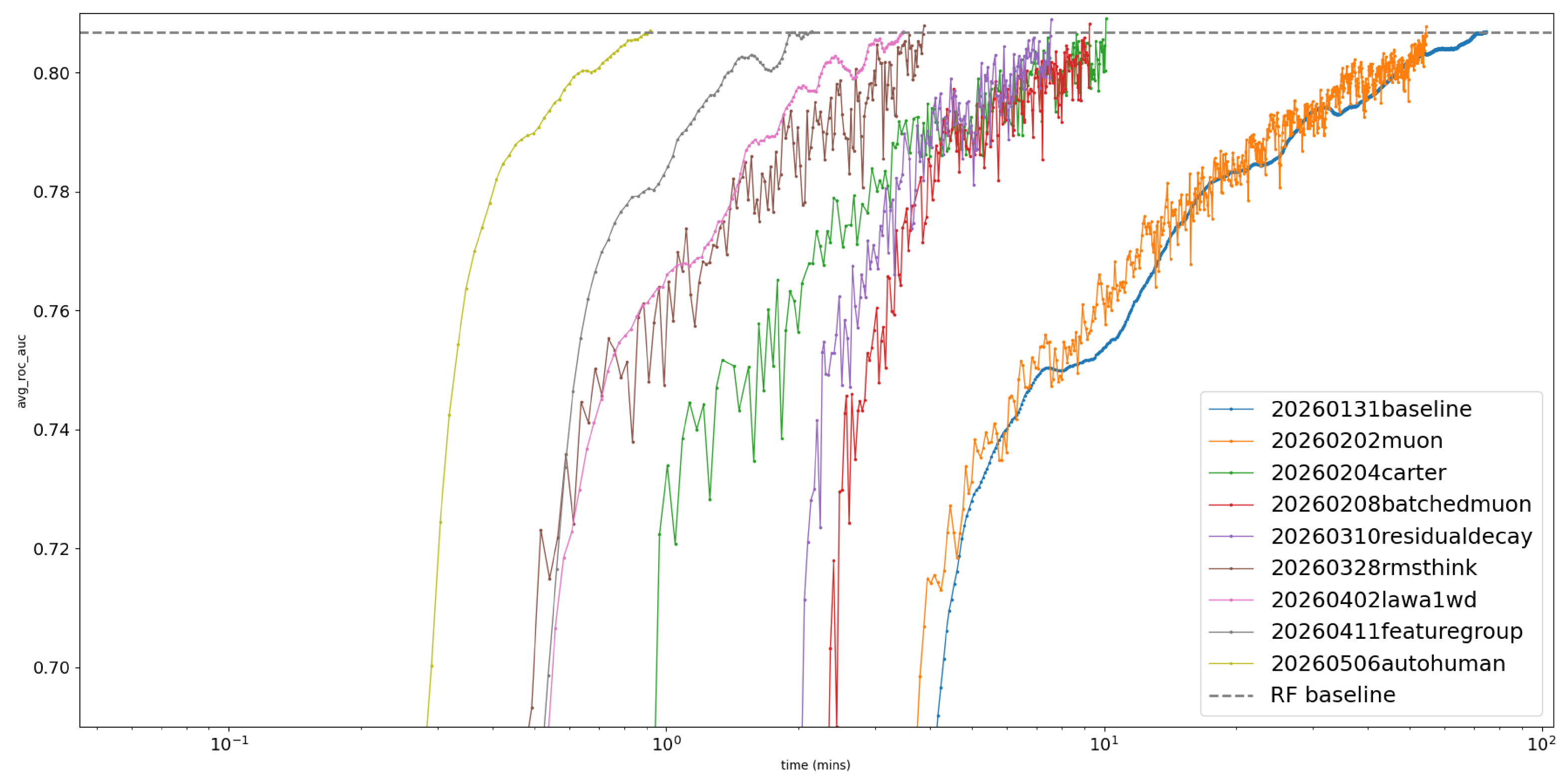
- if >100 features, randomly select 100
- if >1000 rows, randomly select 1000 (stratified by class labels)
- 5-fold StratifiedKFold with shuffling, class labels are encoded with integers per fold
- average binary or one-vs-rest ROC AUC over all datasets

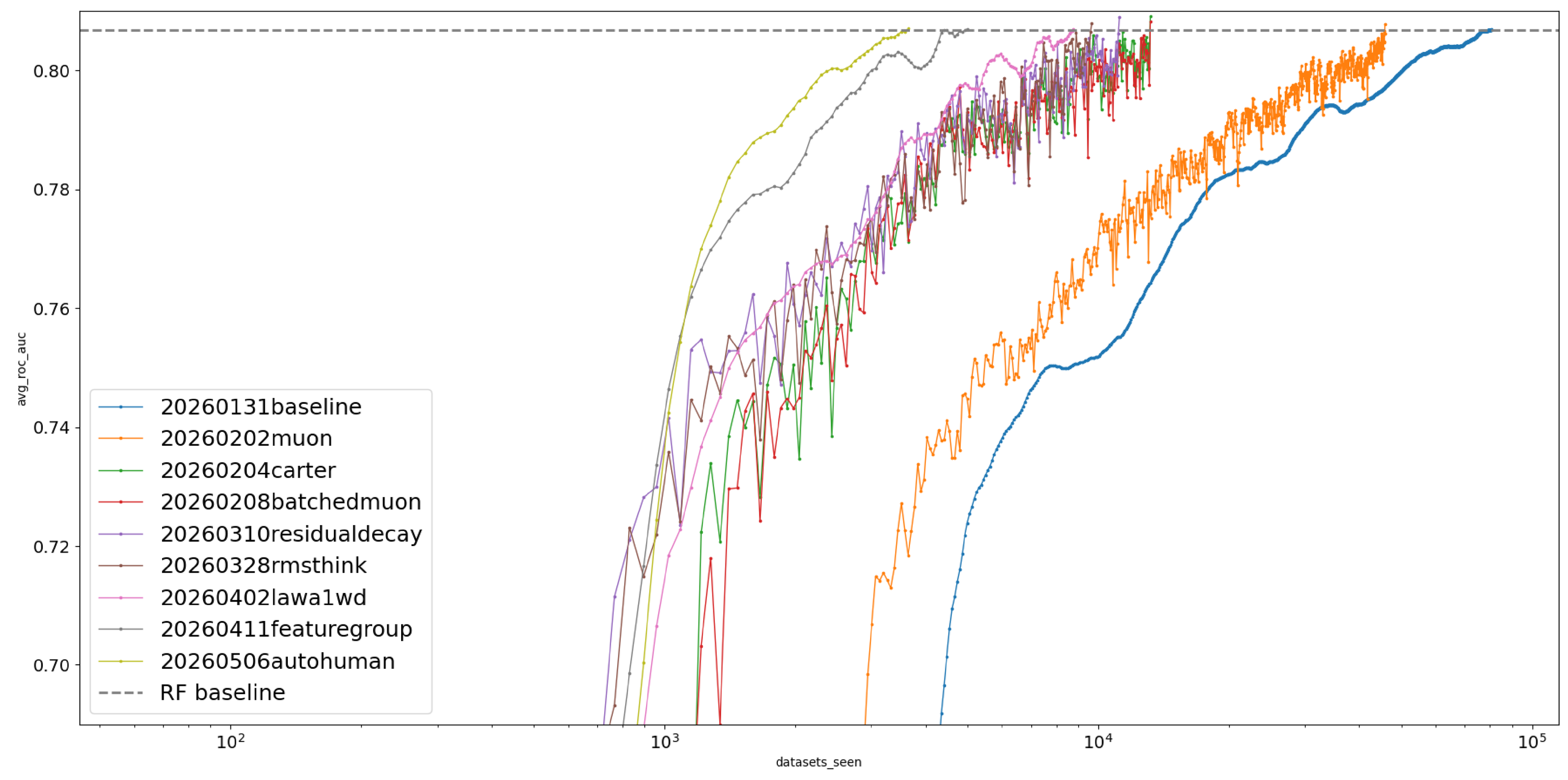
Improvement techniques

This improvement in training speed has been brought about by the following techniques:

- Muon optimizer [3]
- Batched Muon zeropower update for grouped QKV matrices
- Scaled Dot-Product Attention rewrite with explicit QKV
- Pre-norm transformer blocks [4]
- Compile TransformerEncoderLayer forward
- bfloat16 autocast in training and inference
- Set float32 matmul precision to high
- Increase learning rate from 10^{-4} to 10^{-3}
- Increase embedding size from 192 to 256, reduce attention heads from 6 to 4
- Exponential decay of the residual stream across layers
- Lower precision RMSNorm [5]
- Prepend 24 learnable Thinking Rows [6]
- Latest Weight Averaging of 10 checkpoints [7, 8]
- Set Schedule-Free AdamW weight decay to 0.01 [9]
- Repeated feature grouping (group size 5) [10]
- Reduce transformer layers from 6 to 5, increase batch size from 1 to 2
- Add decoupled weight decay 0.1 to Muon, increase momentum from 0.95 to 0.96
- Increase gradient clip from 1.0 to 2.0
- Feed mean of test feature embeddings into output decoder
- Autoresearch-driven HPO and architecture search [11]

Record history





Record time	Description	Contributors
74.32 mins	Baseline	
54.41 mins	Muon optimizer	@borawhocodess
10.10 mins	SDPA, bf16, higher LR, wider embeddings, fewer heads	@carterprince
9.26 mins	Batched Muon, compiled forward	@carterprince
7.57 mins	Exponential decay of residual stream	@borawhocodess
3.88 mins	RMSNorm, Thinking Rows	@borawhocodess
3.48 mins	LAWA, Schedule-Free AdamW WD	@borawhocodess
2.15 mins	Repeated feature grouping	@borawhocodess
0.92 mins	Autoresearch HPO, Muon WD, mean feature pooling	@borawhocodess

Join the race!

Scan to join the speedrun on GitHub
github.com/borawhocodess/modded-nanotabpfn

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